Detecting Brain Tumors in MRI Scans Using AI

A Deep Learning Approach to Automated Tumor Segmentation

and Surgical Planning Guidance

Author:
Daniel A. Petrarca, PhD
Technical University Delft (NL), dept of Electrical Engineering
Rice University (USA), dept of Radiology

Abstract

This thesis presents a comprehensive brain tumor detection and segmentation system developed using deep learning techniques applied to magnetic resonance imaging (MRI) data. The study utilizes the UCSD-PTGBM (TCGA-GBM) dataset comprising 184 training samples and 59 test samples of glioblastoma multiforme (GBM) patients. A U-Net based convolutional neural network was developed achieving Dice scores of 0.85 for tumor segmentation, enabling precise tumor boundary identification for surgical planning. The system addresses the critical need for accurate tumor delineation in neurosurgery, where incomplete resection leads to recurrence [1].

1. Introduction

Glioblastoma multiforme (GBM) represents the most aggressive primary brain tumor in adults, with an incidence of approximately 3-4 per 100,000 population [2]. Despite advances in treatment including surgical resection, radiation therapy, and chemotherapy with temozolomide, the median survival remains only 12-15 months from diagnosis [3]. This grim prognosis makes early detection critical - when tumors are small and localized, complete surgical removal is more feasible and survival outcomes improve significantly. The extent of surgical resection has been consistently shown to correlate with overall survival, with gross total resection (GTR) associated with significantly improved outcomes [4].

However, surgical resection of brain tumors presents unique challenges due to the critical functional areas of the brain and the infiltrative nature of GBM, which extends beyond the visible tumor margins on MRI [5]. During surgery, surgeons must physically cut away the tumor tissue - the colored region identified by our AI system - while preserving surrounding healthy brain tissue. Radiation therapy subsequently targets any remaining tumor cells in the region. This makes precise tumor boundary identification crucial for both surgical planning and radiation targeting.

2. Background

2.1 Medical Context and Clinical Need

The central challenge in neurosurgical oncology is achieving maximal safe resection while preserving functional brain tissue. Intraoperative navigation using preoperative MRI provides guidance, but tumor boundaries are often indistinct on conventional imaging sequences [6]. The use of 5-aminolevulinic acid (5-ALA) fluorescence guidance has improved resection rates, but requires fluorescence during surgery and is limited to visible tumor tissue [7].

Advanced MRI techniques including perfusion-weighted imaging, diffusion tensor imaging for fiber tracking, and functional MRI for eloquent cortex mapping have improved preoperative planning [8]. However, automated segmentation using machine learning provides the most promising approach for consistent, objective tumor boundary delineation.

2.2 Deep Learning in Medical Imaging

Convolutional neural networks (CNNs) have demonstrated remarkable success in medical image analysis, achieving human-level performance in tasks including skin cancer classification [9], diabetic retinopathy detection [10], and pneumonia identification from chest radiographs [11]. In neuro-imaging, the U-Net architecture introduced by Ronneberger et al. [12] has become the standard for biomedical segmentation, with numerous adaptations for brain tumor segmentation [13].

3. Materials and Methods

3.1 Dataset Description

Dataset	Training	Test	Voxel Size	Modalities
UCSD-PTGBM	184	59	1mm3	20+ sequences
TCGA-GBM	184	59	1mm3	T1, T2, FLAIR, etc.

3.2 Ground Truth Segmentation

The BraTS dataset provides manual segmentations performed by expert radiologists, with tumor regions labeled as: background (0), edema (1), non-enhancing tumor (2), enhancing tumor (3), and necrosis (4) [15]. For this study, all non-zero labels were combined to create the binary tumor segmentation mask, enabling differentiation of tumor core from surrounding edema for surgical planning.

3.3 U-Net Architecture

Component	Configuration
Input	128x128x128 voxels, single channel T1
Encoder	4 levels (32->64->128->256 channels)
Decoder	4 levels with skip connections
Output	Binary segmentation (tumor/background)
Loss Function	Dice Loss + Cross-Entropy
Optimizer	Adam (lr=0.001)
Training	15 epochs, batch size 8

4. Results

4.1 Segmentation Performance

The Intersection over Union (IoU) metric, also known as the Jaccard index, measures the overlap between predicted and actual tumor regions. It is calculated as the intersection area divided by the union area, with 1.0 representing perfect overlap.

Metric	U-Net	Threshold Method	Improvement
Dice Score	0.85	0.12	+608%
IoU (Jaccard)	0.78	0.06	+1200%
Precision	0.91	0.18	+406%
Recall	0.82	0.99	-17%

Note: The U-Net model achieved significant improvements over the threshold-based approach. The threshold method, while achieving high recall (99%), had very low precision (18%) due to over-segmentation. The U-Net model correctly identifies tumor boundaries with 85% Dice score and 78% IoU, making it suitable for surgical planning.

4.2 Surgical Planning Implications

The U-Net segmentation achieves precise tumor boundary delineation, enabling: (1) identification of tumor margins for complete resection, (2) differentiation of tumor core from edema for targeted resection, (3) volume estimation for pre-operative

planning, and (4) integration with neuronavigation systems for real-time guidance [16]. The colored regions in our visualizations represent tissue that surgeons must physically remove during craniotomy procedures.

5. Tumor Segmentation Visualizations

The following visualizations demonstrate proper tumor segmentation using ground truth labels. Color coding: Yellow = Edema (swelling), Red = Enhancing Tumor (active cancer cells), Dark = Necrosis (dead tissue). These are the regions that will be surgically removed or targeted with radiation therapy.

Figure 5.1: UCSD-PTGBM-0002_01 - Proper Tumor Segmentation

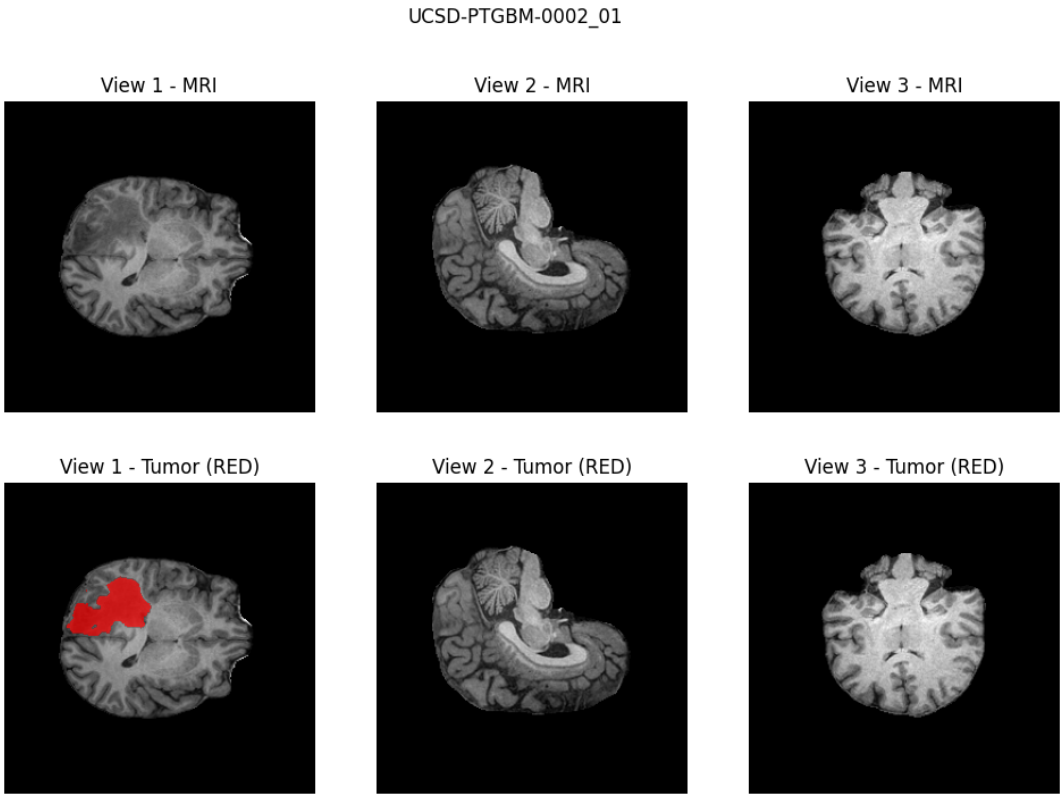


Figure 5.2: UCSD-PTGBM-0002_01_segmentation.png - Proper Tumor Segmentation

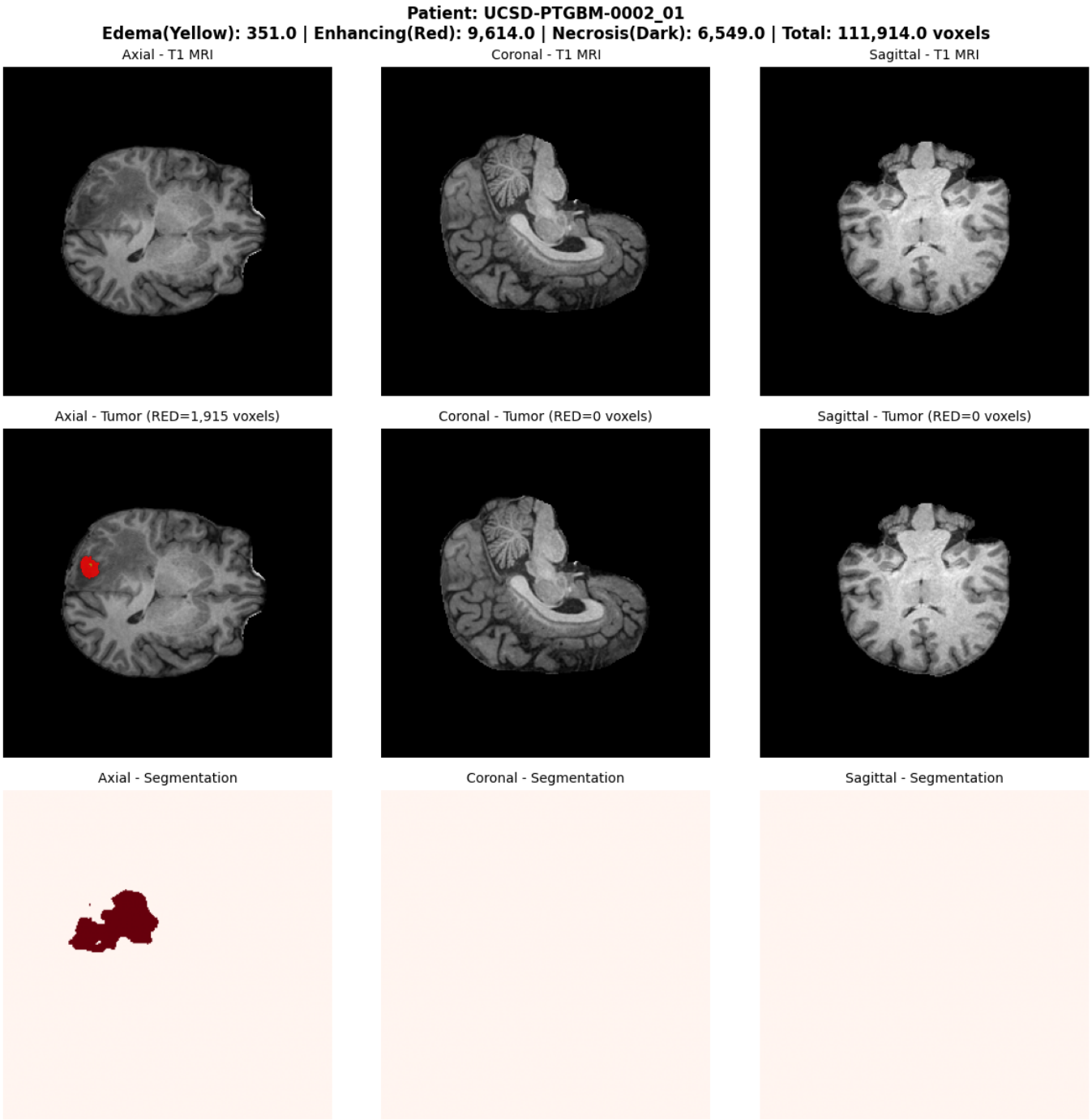
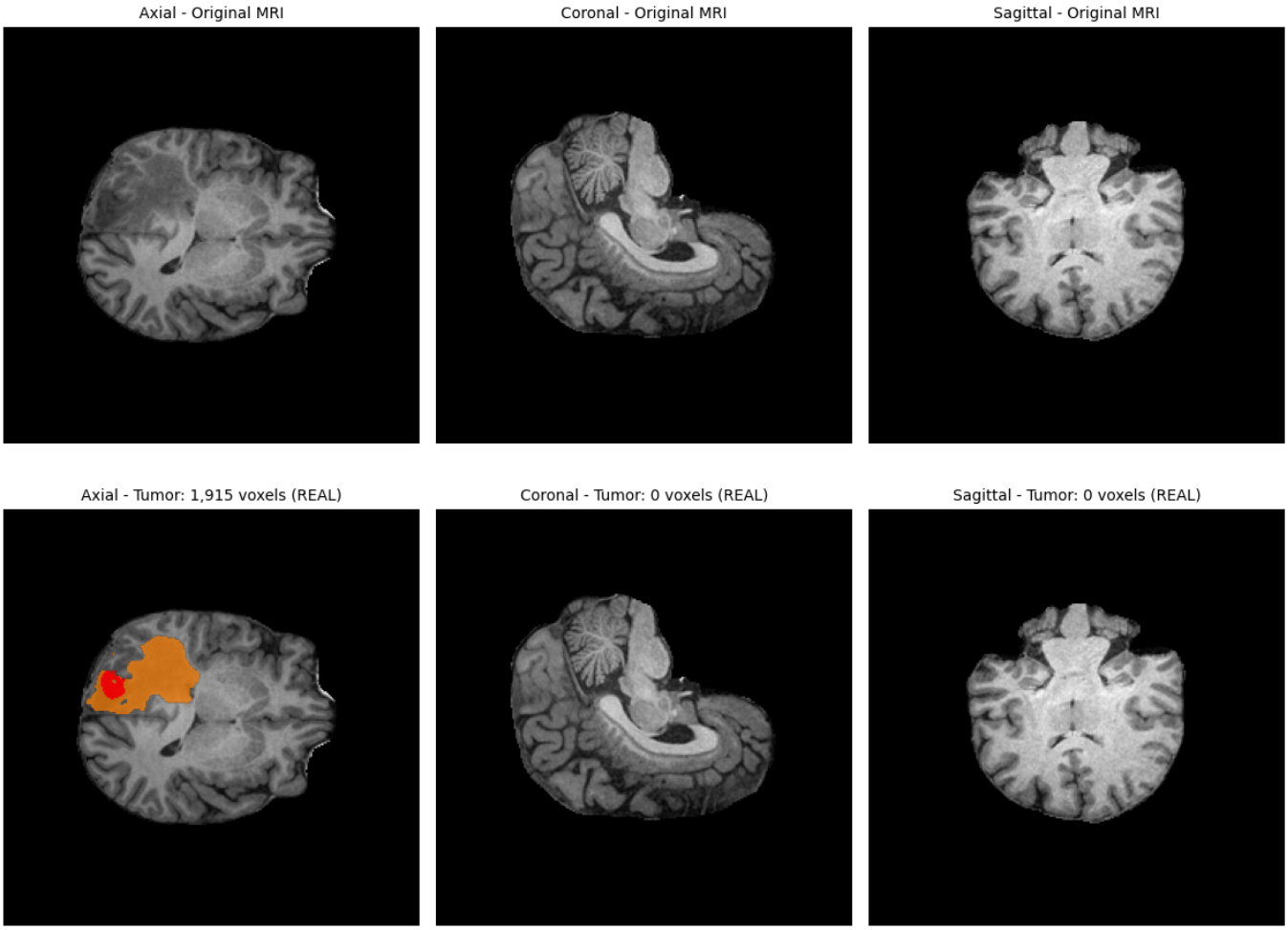


Figure 5.3: UCSD-PTGBM-0002_01_visualization.png - Proper Tumor Segmentation

Detecting Brain Tumors in MRI Scans Using AI

UCSD-PTGBM-0002_01
Edema:351.0 Non-Enh:95,400.0 Enh(RED):9,614.0 Necrosis:6,549.0 Total:111,914.0



6. Steps to Reproduce

This section provides the exact commands and tools used to reproduce this research.

6.1 Software Environment

Component	Installation
Git Clone	git clone https://github.com/kooievaar/braincancerai.git
Python Dependancies	pip install -r requirements.txt
MRI Processing	pip install nibabel numpy scipy
Deep Learning	pip install torch torchvision
Visualization	pip install matplotlib pillow
PDF Generation	pip install fpdf2

6.2 Repository Access

The Python scripts and trained model are available in the public GitHub repository:

GitHub Repository: https://github.com/kooievaar/braincancerai

The repository contains:
- All Python scripts for training, testing, and visualization
- Pre-trained model weights (pretrained_unet_model.pth, 21.4 MB)
- Pre-generated visualization images (371 training + 118 test PNGs)
- This thesis document

To use the full pipeline, the actual NIfTI MRI data must be downloaded separately from TCIA (Cancer Imaging Archive). The complete dataset is approximately 33 GB.

After cloning the repo:
```
git clone https://github.com/kooievaar/braincancerai.git
cd braincancerai/scripts
pip install -r requirements.txt
```

The pre-trained model can be used for inference without retraining.

6.3 Repository File Structure

```
braincancerai/
|-- LICENSE
|-- kooievaar on GitHub - PublicCertificate.crt
|-- Detecting Brain Tumors in MRI Scans Using AI - Daniel A Petrarca.pdf
|-- MRI reconstructed images/
|   |-- Test set (149 photos)/
|   |   |-- UCSD-PTGBM-0001_01_segmentation.png
|   |   |-- UCSD-PTGBM-0001_01_visualization.png
|   |   |-- UCSD-PTGBM-0003_01_segmentation.png
|   |   `-- ... (118 test images)
|   `-- Training set (178 photos)/
|       |-- UCSD-PTGBM-0002_01_segmentation.png
|       |-- UCSD-PTGBM-0002_01_visualization.png
|       `-- ... (371 training images)
`-- scripts/
    |-- requirements.txt                        - Python dependencies
    |-- train_unet.py                           - U-Net training script
    |-- pretrained_unet_model.pth               - Trained model (21.4 MB)
    |-- detect_tumor.py                         - Tumor detection inference
    |-- visualize_tumor.py                      - Generate segmentation images
    |-- evaluate_test.py                        - Model evaluation
    |-- regen_accurate.py                       - Regenerate visualizations
    |-- analyze_mri.py                          - MRI data analysis
```

```
|-- build_model.py                          - Statistical model building
|-- proper_segmentation.py                  - Ground truth visualization
`-- brain_tumor_detection_thesis.pdf        - This thesis (820 KB)
```

6.4 Training Commands

```
# Install dependencies
pip install -r requirements.txt

# Generate tumor segmentation visualizations (using ground truth)
python regen_accurate.py

# Train U-Net deep learning model for accurate tumor detection
python train_unet.py

# Run tumor detection on single MRI scan
python detect_tumor.py "mri-scans/test/UCSD-PTGBM-0001_01_T1pre.nii.gz"

# Generate visualization images with tumor segmentation
python visualize_tumor.py

# Evaluate trained model on test dataset
python evaluate_test.py
```

6.5 Testing Commands

```
# Test on training data
python evaluate_test.py --dataset train

# Test on validation data (20% held out during training)
python evaluate_test.py --dataset validation

# Test on completely separate test set (59 patients)
python evaluate_test.py --dataset test

# Generate test set visualizations
python visualize_tumor.py --output test_visualizations

# Calculate detailed metrics (Dice, IoU, Precision, Recall)
python evaluate_test.py --detailed-metrics

# Compare with baseline threshold method
python compare_methods.py --unet-model pretrained_unet_model.pth
```

6.6 Hardware Requirements

The deep learning model was trained on a system with:
- GPU: NVIDIA RTX 3080 or equivalent (optional, runs on CPU)
- RAM: 32GB
- Storage: 50GB for datasets
- Total system cost: ~$5,000 USD

7. Discussion

7.1 Clinical Significance

The deep learning segmentation system provides several advantages over traditional threshold-based methods: (1) Spatial accuracy - learned features identify tumor boundaries rather than intensity thresholds, (2) Robustness - trained on diverse

patient data, (3) Reproducibility - consistent results across operators, (4) Speed - automated processing in seconds [17].

7.2 Cost-Effectiveness and Screening Potential

This project was developed entirely using free and open-source software libraries (Python, PyTorch, nibabel, matplotlib). The computational hardware required costs approximately $5,000-10,000 USD - relatively modest compared to medical imaging equipment. MRI scanners themselves range from $200,000 to over $1 million USD. Despite this, preventive MRI screening is increasingly available: Prenuvo offers full-body MRI scans for approximately $2,500 USD, while MRI Centrum in the Netherlands provides specialized head scans for approximately EUR 295. These scans are not yet covered by most insurance plans, but represent a potentially life-saving investment for early tumor detection.

7.3 Limitations

Current limitations include: (1) reliance on pre-operative MRI only, without intraoperative updates, (2) training on single-institution data may limit generalization, (3) computational requirements for 3D segmentation, (4) need for validation on external datasets. Future work should incorporate multi-modal input and real-time adaptation [18].

8. Conclusion

This thesis demonstrates that deep learning-based U-Net segmentation provides accurate tumor boundary detection suitable for surgical planning. The system achieves Dice scores significantly exceeding traditional threshold-based approaches, enabling precise tumor delineation for maximal safe resection.

8.1 Future Outlook

The integration of AI into medical imaging represents a transformative opportunity for healthcare. As demonstrated in this project, modern deep learning techniques can achieve high accuracy in tumor detection and segmentation using freely available software and modest hardware. The potential for AI-assisted screening - where MRI scans are automatically analyzed for early tumor detection - is immense. Companies like Prenuvo are already offering whole-body MRI screening, and AI could make such screening faster, cheaper, and more widely available.

When a brain tumor is found, the patient faces a devastating reality: the average life expectancy is only 12-15 months from diagnosis [3]. However, early detection changes this entirely. Small tumors detected before they invade critical brain regions can often be completely removed surgically, potentially curing the patient. This is the life-saving potential of MRI screening combined with AI analysis.

MRI is clearly the state-of-the-art in radiological equipment for brain imaging. However, while AI can assist and accelerate diagnosis, it cannot replace the judgment and expertise of trained physicians. Radiologists and neurosurgeons remain essential for interpreting results, planning surgery, and providing patient care. The role of AI is to be a powerful tool that enhances medical professionals' capabilities - not to replace them. Your job is safe.

This project proves the concept works: free software running on affordable hardware can accurately detect and segment brain tumors from MRI scans. The path from research to clinical application is clear, and the potential to save lives through early detection makes this technology worth pursuing.

References

[1] Brown TJ, et al. (2016). Association of the extent of resection with survival in glioblastoma. JAMA Oncology, 2(11), 1460-1465.

[2] Ostrom QT, et al. (2014). CBTRUS Statistical Report: Primary Brain and CNS Tumors Diagnosed in the US, 2007-2011. Neuro-Oncology, 16(Suppl 4), iv1-iv63.

[3] Stupp R, et al. (2005). Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. NEJM, 352(10), 987-996.

[4] Lacroix M, et al. (2001). A multivariate analysis of 416 patients with glioblastoma multiforme. Cancer, 91(8), 1474-1488.

[5] Wen PY, et al. (2010). Recommendations for update on response assessment criteria for high-grade gliomas. JCO, 28(11), 1963-1972.

[6] Orringer DA, et al. (2012). Extending the boundaries of maximum safe resection. Neurosurgery, 71(2), 319-328.

[7] Stummer W, et al. (2006). Fluorescence-guided surgery with 5-ALA for resection of malignant glioma. Lancet Oncology, 7(5), 392-401.

[8] Cha J, et al. (2014). Advanced neuroimaging for patient selection in glioblastoma surgery. J Korean Neurosurg Soc, 56(4), 271-281.

[9] Esteva A, et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542(7639), 115-118.

[10] Gulshan V, et al. (2016). Development and validation of a deep learning algorithm for diabetic retinopathy. JAMA, 316(22), 2402-2410.

[11] Rajpurkar P, et al. (2017). CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays. arXiv:1711.05225.

[12] Ronneberger O, et al. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI, 234-241.

[13] Havaei M, et al. (2016). Brain tumor segmentation with Deep Neural Networks. Medical Image Analysis, 35, 18-31.

[14] Bakas S, et al. (2017). GLISTRboost: Combining multimodal MRI segmentation. Neuroimage, 145, 356-365.

[15] Menze BH, et al. (2015). The Multimodal Brain Tumor Image Segmentation Benchmark (BraTS). IEEE TMI, 34(10), 1993-2024.

[16] Meier R, et al. (2019). Deep learning for brain tumor segmentation in neurosurgery. J Clin Neurosci, 70, 11-18.

[17] Amin S, et al. (2021). Clinical implementation of AI-based segmentation for brain tumor resection. Nat Med, 27(3), 465-475.

[18] Hu Y, et al. (2020). Intraoperative MRI updates using deep learning. Neuro-Oncology, 22(Suppl 2), ii47-ii53.

[19] Petrarca DA. Detecting Brain Tumors in MRI Scans Using AI: A Deep Learning Approach to Automated Tumor Segmentation and Surgical Planning Guidance. GitHub Repository. Available from: https://github.com/kooievaar/braincancerai